CURRICULUM VITAE SHIWEI LIU

BACKGROUND

Name	Shiwei Liu
University	The University of Texas at Austin
Web Page	`https://shiweiliuiiiiiii.github.io/`
Address	800 W 38TH ST APT2304, Austin, Texas, US
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EDUCATIONAL HISTORY

Institute for Foundations of Machine Learning (IFML), US *September 2022 - Present*
Postdoc
Department: IFML & Computer Science
Supervisors: Dr. Atlas Wang

Eindhoven University of Technology (TU/e), The Netherlands *March 2018 - April 2022*
Doctor of Philosophy (Ph.D.)
Thesis: Sparse Neural Network Training with In-Time Over-Parameterization
Department: Mathematics and Computer Science
Promotors: Prof. Dr. Mykola Pechenizkiy; Dr. Decebal Constantin Mocanu

Harbin Institute of Technology, China *September 2015 - July 2017*
Master of Science (M.Sc.)
Department: Mechanical Engineering and Automation
Advisor: Prof. Dr. Ai-Guo Wu

North University of China, China *September 2011 - July 2015*
Bachelor of Engineering (B.Eng.)
Department: Mechanical Engineering and Automation
Advisor: Dr. Zhongqing Wang

RESEARCH INTERESTS

Understanding Deep Neural Networks From A Scientific Perspective. Despite the ever-strong results achieved by deep neural networks, how to fully understand the behavior of deep learning remains mysterious, resulting in potential risks to our society and industry. I study and understand the behavior of deep learning from a scientific perspective, by conducting empirical experiments in practice.

Designing Efficient Sparse Neural Networks and Training Recipes. Costs associated with training and deploying modern deep neural networks have increased exponentially, beyond the reach of hardware development. I design efficient neural networks by inducing sparsity, improving the efficiency of deep neural networks without sacrificing performance.

AWARDS AND HONOURS

Best Paper Award, Learning on Graphs Conference (LoG 2022)	*11/2022*
AAAI-23 Travel Scholarship	*12/2022*
IFML Postdoctoral Fellowship, US	*9/2022*
Cum Laude (distinguished Ph.D. thesis), TU/e, acceptance rate: 5% in University	*4/2022*
Outstanding Intern, JD Explore Academy, 2% in company	*10/2021*
China National Scholarship, 0.2% in China	*10/2016*

The First Prize Scholarship *07/2016*
The First Prize Scholarship *07/2017*
The Outstanding Graduate *07/2015*

SELECTED PUBLICATIONS

[**1**] Tianjin Huang, Tianlong Chen, Meng Fang, Vlado Menkovski, Jiaxu Zhao, Lu Yin, Yulong Pei, Decebal Constantin Mocanu, Zhangyang Wang, Mykola Pechenizkiy, **Shiwei Liu**. *You Can Have Better Graph Neural Networks by Not Training Weights at All: Finding Untrained Graph Tickets.* Learning on Graphs Conference (LoG), 2022. **Best Paper Award**.

[**2**] **Shiwei Liu**, Lu Yin, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *Do We Actually Need Dense Over-Parameterization? In-Time Over-Parameterization in Sparse Training.* The Thirty-Eighth International Conference on Machine Learning (ICML), PMLR, 2021.

[**3**] **Shiwei Liu***, Tianlong Chen*, Zhenyu Zhang, Xuxi Chen, Tianjin Huang, AJAY KUMAR JAISWAL, and Zhangyang Wang. *Sparsity May Cry: Let Us Fail (Current) Sparse Neural Networks Together!.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-25% Spotlight**.

4] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Xuxi Chen, Qiao Xiao, Boqian Wu, Mykola Pechenizkiy, Decebal Mocanu, and Zhangyang Wang. *More Convnets in the 2020s: Scaling Up Kernels Beyond 51x51 Using Sparsity.* International Conference on Learning Representations (ICLR), 2023.

[**5**] Duc N.M Hoang, **Shiwei Liu**, Radu Marculescu, Zhangyang Wang. *Revisiting Pruning at Initialization Through the Lens of Ramanujan Graph.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-5% Spotlight**.

TECHNICAL WORK EXPERIENCE

Internship: JD Explore Academy, China *June 2021 - October 2021*
Collaborators: Li Shen, Dacheng Tao
Responsibility: Developing efficient and accurate sparse deep neural algorithms and systems; leveraging deep learning and machine learning on big data resources.

Internship: Chinese University of Hong Kong (CUHK), China *May 2016 - August 2016*
Responsibility: Modifying and applying SLAM into the turtlebot and controlling it to its destination automatically.

SERVICE

2023: Area Chair: ICIP 2023
Conference Reviewer: ICML, CVPR, ICCV, UAI

2022: Area Chair: ICIP 2022
Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, AAAI.
Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence.

2021: Conference Reviewer: NeurIPS, ICLR, ICML, AISTATS.
Journal Reviewer: IEEE Transactions on Evolutionary Computation.

2020: Conference Reviewer: ECMLPKDD, IDA.
Journal Reviewer: ACM Transactions On Intelligent Systems And Technology.

INVITED TALKS & TUTORIALS

Tutorial: Sparse Neural Networks Training in ECMLPKDD 2022, Sep 2022.
Talk: Fantastic Sparse Masks and Where to Find Them, Cong (Callie) Haos group, Georgia Institute of Technology, Mar 2023.
Talk: Fantastic Sparse Masks and Where to Find Them, IFML, UT Austin, Jan 2023.
Talk: Sparse Neural Network Training, Dongkuan Xu's Group Meeting, NC State, Nov 2022.
Talk: More Convnets in the 2020s: Scaling up Kernels beyond 51x51, RICE University, Sep 2022.
Talk: More Convnets in the 2020s: Scaling up Kernels beyond 51x51, TechBeat, Sep 2022.
Talk: Sparse Training in Model Compression and Scaling, Zhidx (AI XinQingNian), Nov 2022.

PEER REVIEWED PUBLICATIONS

Publication Summary

Overall: 36 papers, **18** first-author papers, **4** journal papers, and **14** preprints.
Impact: 321 citations, h-index: **10**, i10-index: **10** (as of March 12th 2023).

Conference Publications

[**1**] Shiwei Liu*, Tianlong Chen*, Zhenyu Zhang, Xuxi Chen, Tianjin Huang, AJAY KUMAR JAISWAL, and Zhangyang Wang. *Sparsity May Cry: Let Us Fail (Current) Sparse Neural Networks Together!.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-25% Spotlight**.

[**2**] Shiwei Liu, Tianlong Chen, Xiaohan Chen, Xuxi Chen, Qiao Xiao, Boqian Wu, Mykola Pechenizkiy, Decebal Mocanu, and Zhangyang Wang. *More Convnets in the 2020s: Scaling Up Kernels Beyond 51×51 Using Sparsity.* International Conference on Learning Representations (ICLR), 2023.

[**3**] Duc N.M Hoang, Shiwei Liu, Radu Marculescu, Zhangyang Wang. *Revisiting Pruning at Initialization Through the Lens of Ramanujan Graph.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-5% Spotlight**.

[**4**] Tianlong Chen, Zhenyu Zhang, AJAY KUMAR JAISWAL, Shiwei Liu, Zhangyang Wang *Sparse MoE with Random Routing as the New Dropout: Training Bigger and Self-Scalable Models.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-25% Spotlight**.

[**5**] Lu Yin*, Shiwei Liu*, Fang Meng, Tianjin Huang, Vlado Menkovski, Mykola Pechenizkiy. *Lottery Pools: Winning More by Interpolating Tickets without Increasing Training or Inference Cost.* Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), 2023.

[**6**] Tianjin Huang, Tianlong Chen, Meng Fang, Vlado Menkovski, Jiaxu Zhao, Lu Yin, Yulong Pei, Decebal Constantin Mocanu, Zhangyang Wang, Mykola Pechenizkiy, Shiwei Liu. *You Can Have Better Graph Neural Networks by Not Training Weights at All: Finding Untrained Graph Tickets.* Learning on Graphs Conference (LoG), 2022. **Best Paper Award**.

[**7**] Qiao Xiao, Boqian Wu, Yu Zhang, Shiwei Liu, Mykola Pechenizkiy, Elena Mocanu, Decebal Constantin Mocanu. *Dynamic Sparse Network for Time Series Classification: Learning What to See.* The Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022.

[**8**] Lu Yin, Vlado Menkovski, Meng Fang, Tianjin Huang, Yulong Pei, Mykola Pechenizkiy, Shiwei Liu. *Superposing Many Tickets into One: A Performance Booster for Sparse Neural Network Training.* The 38th Conference on Uncertainty in Artificial Intelligence (UAI), 2022.

[**9**] Shiwei Liu, Tianlong Chen, Xiaohan Chen, Li Shen, Decebal Constantin Mocanu, Zhangyang Wang, and Mykola Pechenizkiy. *The Unreasonable Effectiveness of Random Pruning: Return of the Most*

Naive Baseline for Sparse Training. International Conference on Learning Representations (ICLR), 2022.

[**10**] Shiwei Liu, Tianlong Chen, Zahra Atashgahi, Xiaohan Chen, Ghada Sokar, Elena Mocanu, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Deep Ensembling with No Overhead for either Training or Testing: The All-Round Blessings of Dynamic Sparsity.* International Conference on Learning Representations (ICLR), 2022.

[**11**] Shiwei Liu, Tianlong Chen, Xiaohan Chen, Zahra Atashgahi, Lu Yin, Huanyu Kou, Li Shen, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Sparse Training via Boosting Pruning Plasticity with Neuroregeneration.* The Thirty-fifth Conference on Neural Information Processing Systems (NeurIPS), 2021.

[**12**] Shiwei Liu, Lu Yin, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *Do We Actually Need Dense Over-Parameterization? In-Time Over-Parameterization in Sparse Training.* The Thirty-eighth International Conference on Machine Learning (ICML), PMLR, 2021.

[**13**] Shiwei Liu, Decebal Constantin Mocanu, Yulong Pei, and Mykola Pechenizkiy. *Selfish sparse RNN training.* The Thirty-eighth International Conference on Machine Learning (ICML), PMLR, 2021.

[**14**] Shiwei Liu, Tim Van der Lee, Anil Yaman, Zahra Atashgahi, Davide Ferraro, Ghada Sokar, Mykola Pechenizkiy, and Decebal Constantin Mocanu. *Topological Insights into Sparse Neural Networks.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), Ghent, Belgium, 2020.

[**15**] Shiwei Liu. *Learning Sparse Neural Networks for Better Generalization.* 29th International Joint Conference on Artificial Intelligence-17th Pacific Rim International Conference on Artificial Intelligence (IJCAI), 2020. Doctoral Consortium.

[**16**] Shiwei Liu, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *On improving deep learning generalization with adaptive sparse connectivity.* The Workshop of Understanding and Improving Generalization in Deep Learning, The Thirty-sixth International Conference on Machine Learning (ICML), PMLR, 2019.

[**17**] Yan, Fulong, Shiwei Liu, and Nicola Calabretta. *Network Performance Optimization with Real Time Traffic Prediction in Data Center Network.* 2020 European Conference on Optical Communications (ECOC). IEEE, 2020.

Journal Publications

[**18**] Zahra Atashgahi, Joost Pieterse, Shiwei Liu, Decebal Constantin Mocanu, Raymond Veldhuis, Mykola Pechenizkiy. *A Brain-inspired Algorithm for Training Highly Sparse Neural Networks.* Machine Learning Journal (ECML-PKDD 2022 journal track).

[**19**] Shiwei Liu, Decebal Constantin Mocanu, Amarsagar Reddy Ramapuram Matavalam, Yulong Pei, and Mykola Pechenizkiy. (2021). *Sparse evolutionary deep learning with over one million artificial neurons on commodity hardware.* Neural Computing and Applications, 1-16.

[**20**] Shiwei Liu, Iftitahu Nimah, Vlado Menkovski, Decebal Constantin Mocanu, and Mykola Pechenizkiy. (2021). *Efficient and effective training of sparse recurrent neural networks.* Neural Computing and Applications, 1-12.

[**21**] Zahra Atashgahi, Xuhao Zhang, Neil Kichler, Shiwei Liu, Lu Yin, Mykola Pechenizkiy, Raymond Veldhuis, Decebal Constantin Mocanu. *Feature Selection with Neuron Evolution.*, TMLR. 2023.

Preprint

[**22**] Shiwei Liu, and Zhangyang Wang. *Ten Lessons We Have Learned in the New "Sparseland": A Short Handbook for Sparse Neural Network Researchers.*

[**23**] Shiwei Liu, Yuesong Tian, Tianlong Chen, Li Shen. *Don't Be So Dense: Sparse-to-Sparse GAN Training Without Sacrificing Performance.*

[**24**] Shiwei Liu, Tianjin Huang, Tianlong Chen, Zhangyang Wang. *The Counterattack of CNNs in Self-Supervised Learning: Larger Kernel Size might be All You Need.*

[**25**] Tiansheng Huang, Shiwei Liu, Li Shen, Fengxiang He, Weiwei Lin, Dacheng Tao. *Achieving Personalized Federated Learning with Sparse Local Models.*

[**26**] Tianjin Huang, Lu Yin, Zhenyu Zhang, Li Shen, Meng Fang, Mykola Pechenizkiy, Zhangyang Wang, Shiwei Liu. *Are Large Kernels Better Teachers than Transformers for ConvNets?.* 2023.

[**27**] Lu Yin, Gen Li, Meng Fang, Li Shen, Tianjin Huang, Zhangyang Wang, Vlado Menkovski, Xiaolong Ma, Mykola Pechenizkiy, Shiwei Liu. Dynamic Sparsity Is Channel-Level Sparsity Learner.

[**28**] AJAY KUMAR JAISWAL, Shiwei Liu, Tianlong Chen, Ying Ding, Zhangyang Wang. *Graph Ladling: Embarrassingly Scalable and Efficient training of Powerful GNNs via Data-Centric Model Averaging.*

[**29**] AJAY KUMAR JAISWAL, Shiwei Liu, Tianlong Chen, Ying Ding, Zhangyang Wang. *Instant Soup: Cheap Pruning Ensembles in A Single Pass Can Draw Lottery Tickets from Large Models.*

[**30**] Ruisi Cai, Xiaohan Chen, Shiwei Liu, Jayanth Srinivasa, Myungjin Lee, Ramana Rao Kompella, Zhangyang Wang. *Scaling Federated Learning under Data and Task Heterogeneity: A Pilot Study.*

[**31**] Hoang Pham, The-Anh Ta, Shiwei Liu, Lichuan Xiang, Dung D. Le, Hongkai Wen, Long Tran-Thanh. *Towards Data-Agnostic Pruning At Initialization: What Makes a Good Sparse Mask?*

[**32**] Haoyu Ma, Chengming Zhang, Lizhi Xiang, Xiaolong Ma, Geng Yuan, Wenkai Zhang, Shiwei Liu, Tianlong Chen, Dingwen Tao, Yanzhi Wang, Zhangyang Wang, Xiaohui Xie. *HRBP: Hardware-friendly Regrouping towards Block-wise Pruning for Sparse Training.*

[**33**] Yin Nan, Li Shen, Mengzhu Wang, Shiwei Liu, Chong Chen, Xian-Sheng Hua, Xiao Luo. *SPORT: A Subgraph Perspective on Graph Classification with Label Noise.*

[**34**] Enneng Yang, Li Shen, Zhenyi Wang, Shiwei Liu, Guibing Guo, Xingwei Wang. *Data Augmented Flatness-aware Gradient Projection for Continual Learning.*

[**35**] Zhangheng Li, Shiwei Liu, Tianlong Chen, AJAY JAISWAL, Zhenyu Zhang, Dilin Wang, Raghuraman Krishnamoorthi, Shiyu Chang, Zhangyang Wang. *Sparse Cocktail: Co-Training Many Sparsity Patterns and Ratios at Once.*

Ph.D. Thesis

[**36**] Shiwei Liu. Sparse Neural Network Training with In-Time Over-Parameterization. The Eindhoven University of Technology. 2022.

MENTORING EXPERIENCE

Lu Yin, Ph.D. at Eindhoven University of Technology
Project I: Sparse Neural Network Merging (UAI 2022)
Project II: Mergability of Lottery Ticket Hypothesis (AAAI 2023)
Project III: Channel-Wise Dynamic Sparse Training (Under Review)

Tianjin Huang, Ph.D. at Eindhoven University of Technology
Project I: Untrained GNN Tickets (LoG 2022 Best Paper Award)
Project II: Optimization Trajectory Refining for Robustness (Under Review)
Project III: Large-Kernel Distillation (Under Review)

AJAY KUMAR JAISWAL, Ph.D. student at University of Texas at Austin
Project I: Cheap Tickets Ensemble for Large Models
Project II: Scalable and Efficient Training of GNNs via Data-Centric Model Averaging

Zhenyu Zhang, Ph.D. Student at University of Texas at Austin
Project I: Model Fusion

Zhangheng Li, Ph.D. Student at University of Texas at Austin
Project I: Sparse Co-training Many Sparsity Patterns and Ratios at Once

Ruisi Cai, Ph.D. Student at University of Texas at Austin
Project I: Scaling Federated Learning under Data and Task Heterogeneity

Qiao Xiao, Ph.D. Student at Eindhoven University of Technology
Project I: 1D Large Kernel for Time Series Classification

Boqian Wu, Ph.D. Student at University of Twente
Project I: Dynamic Sparsity for 3D Medical Image Segmentation

Tiansheng Huang, Master Student at **South China University of Technology**
Project I: Personalized Federated Learning with Sparse Local Models